Technische Universität Braunschweig

Decision Support
Institut für Wirtschaftsinformatik

Data Driven Decision Making

Lecture 1 – Overview

Contact „Data Driven Decision Making“

Lecturer

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Website

https://www.tu-braunschweig.de/winfo

https://www.tu-braunschweig.de/winfo/teaching, e.g. exam dates

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Master @ Decision Support

Orientierung Decision Support (5 LP)

- Intelligent Data Analysis (WS)
- Planning for Mobility and Transportation Purposes (WS)

→

Spezialisierung Decision Support (5 LP)

- Data Driven Decision Making (SS)
- Exercise in Data Driven Decision Making (SS)

↓

- Master seminar
- Master thesis (6 months)

In addition: Interested students of all disciplines, especially mechanical engineering students (NT subject, 3 LP)

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About the Lecture

- **Data Driven Decision Making**
- Previously *Informationssysteme für Mobilitätsanwendungen*
- … before *Verkehrsinformationssysteme*

- Shift from depiction of information system to automated decisions in information systems
- Inline with state-of-the-art digital business models like UBER, LIME, AMAZON etc.
- Builds on PMT (focusing on decisions) and IDA (focusing on data analytics)

- With respect to coursework, 3DM represents “Spezialisierung Decision Support” together with 3DM exercise

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Goals

- Understanding stochasticity and dynamism
- Modeling stochastic dynamic decision problems
- Overview over methodology (functionality, strengths, weaknesses, etc.)
- Insight about relevant problems in the digital economy

- Pass exam (60min written exam)

- Preparation for master thesis (and PhD)

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DDDM Schedule

Lecture	Topic
1	Overview
2	Information Modeling and Decision Modeling
3	Stochasticity
4	Dynamism
5	Markov Decision Process
6	Approximate Dynamic Programming
7	Look-ahead Methods
8	Value Function Approximation
9	Analytics Issues in Approximate Dynamic Programming
10	Combined Methods
11	Platform-based Food Delivery

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Literature

- Work by Warren Powell (https://castlelab.princeton.edu/)

- Recent joint work at TU Braunschweig with Marlin Ulmer and others
- Adaptation and development with respect to transportation and mobility

- Ulmer et. al (2020) On modeling stochastic dynamic vehicle routing problems, https://www.sciencedirect.com/science/article/pii/S219243762030008X

- Ulmer et al. (2021) Stochastic dynamic vehicle routing in the light of prescriptive analytics: A review, https://www.sciencedirect.com/science/article/pii/S0377221721006093

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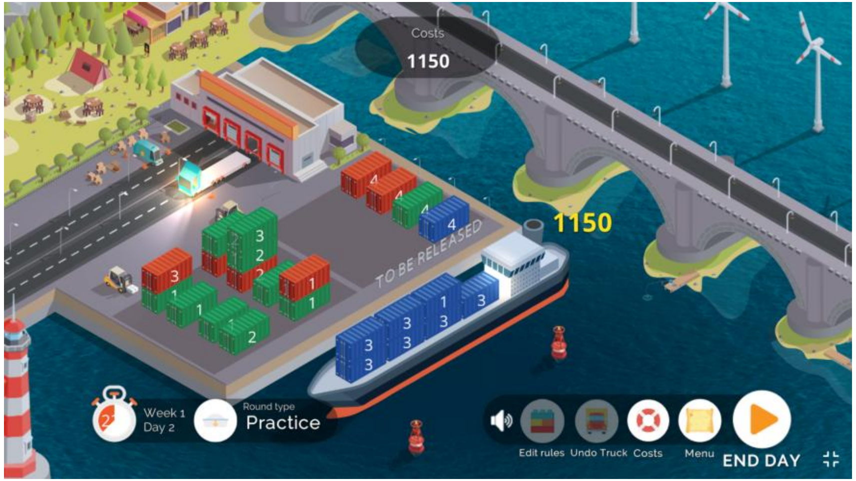
About the Exercise

- “Studienleistung” for “Spezialisierung Decision Support” (2.5 LP)

- Practice lecture content in the serious gaming application “Trucks and Barges”
  - Simulates decision situation of a freight forwarder who can ship containers from a terminal using different transport modes on consecutive days



- Organization:
  - 5 dates with content building on each other (online)
  - Homework assignments
  - StudIP: “Data Driven Decision Making - Übung”

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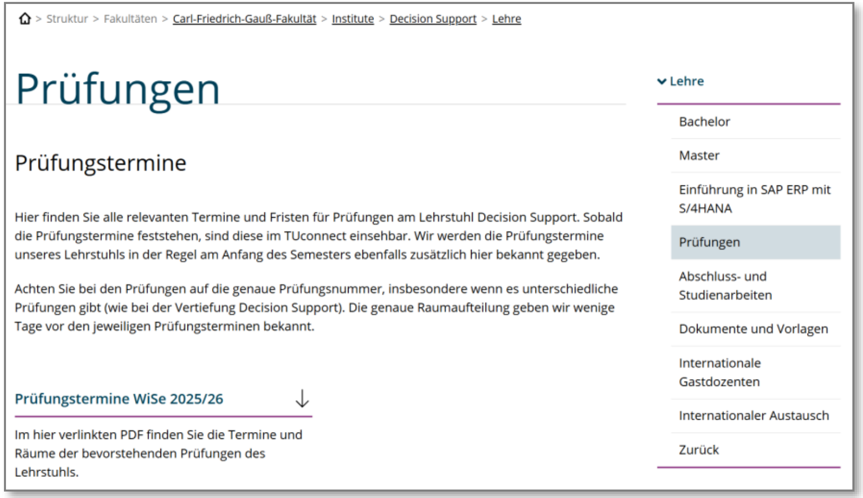
Information on exams

You can find the locations and times of all exams offered by our department on our website:
https://www.tu-braunschweig.de/winfo/teaching/pruefungen

The information is updated regularly.

Exam dates are announced well in advance, while exam rooms are assigned shortly before the exams.

Please only email us if the information you need is not on our website.



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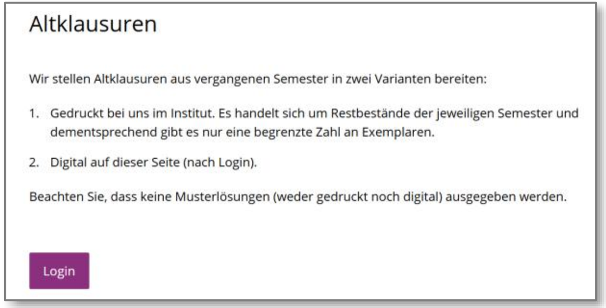
Altklausuren

You can find printed copies of past exams from recent semesters directly at our institute. These are leftover copies, so there are only a limited number available. You can also find past exams on our website: https://www.tu-braunschweig.de/winfo/teaching/pruefungen



However, simply working through past exams is not sufficient preparation for the exam and does not guarantee a good grade. All topics covered in the lecture are relevant to the exam. While the types of questions on the exam are similar each semester, the content may vary, and new questions that do not appear in any of the past exams may also be included. No sample solutions for the past exams will be provided.

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Agenda

- **Motivation**

- Operations Research & Business Analytics (Recap)

- Modeling and Outline of the Lectures

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Motivation

- New and changing business concepts.
  faster, more individually, real-time
- Examples:
  - Instant delivery
  - Production on demand/made to order
  - Appointment scheduling
  - etc.

- Challenging planning and control
- Two common themes:
  - Stochasticity: Information changes over time
  - Dynamism: Updated plans under the light of information change

→ Stochastic Dynamic Decision Processes

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Examples: Transportation

amazon.de


gizmodo.org

lebensmittelzeitung.de


Handelsblatt.de

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Examples: Mobility and Services

www.wnyc.org

wikimedia.or


brk.de

McKinsey


housecallsok.com


schindler.com

ecomento.tv

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Summary

- Service processes over time
- New information (demand, requests, traffic, drivers, battery levels, requirements, etc.)
- Reactions and updates of planning required

- Reactions might be too late (vehicle stuck in traffic, no drivers around to deliver meals, drone crashes, etc.)
- Anticipation of an expected future required
- Aim at flexibility with respect to a future system state

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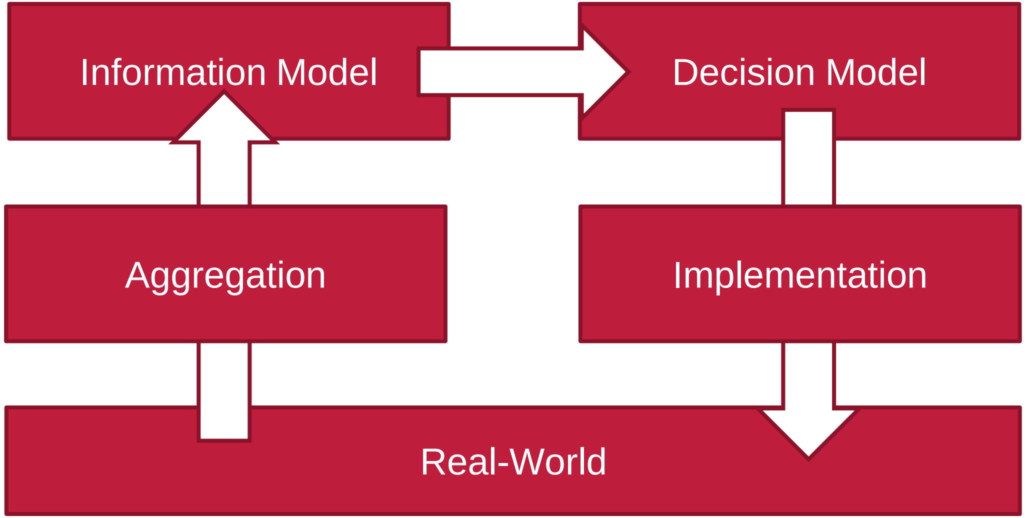
Decision making

- **Real-World:** Actions are performed and resulting states can be observed
- **Aggregation:** Transfers real-world data into dimensions of information model
- **Information Model:** Depicts information in a condensed way
- **Decision Model:** Defines decision space and evaluates decision
- **Implementation:** Translates solution of the decision model into real-world actions



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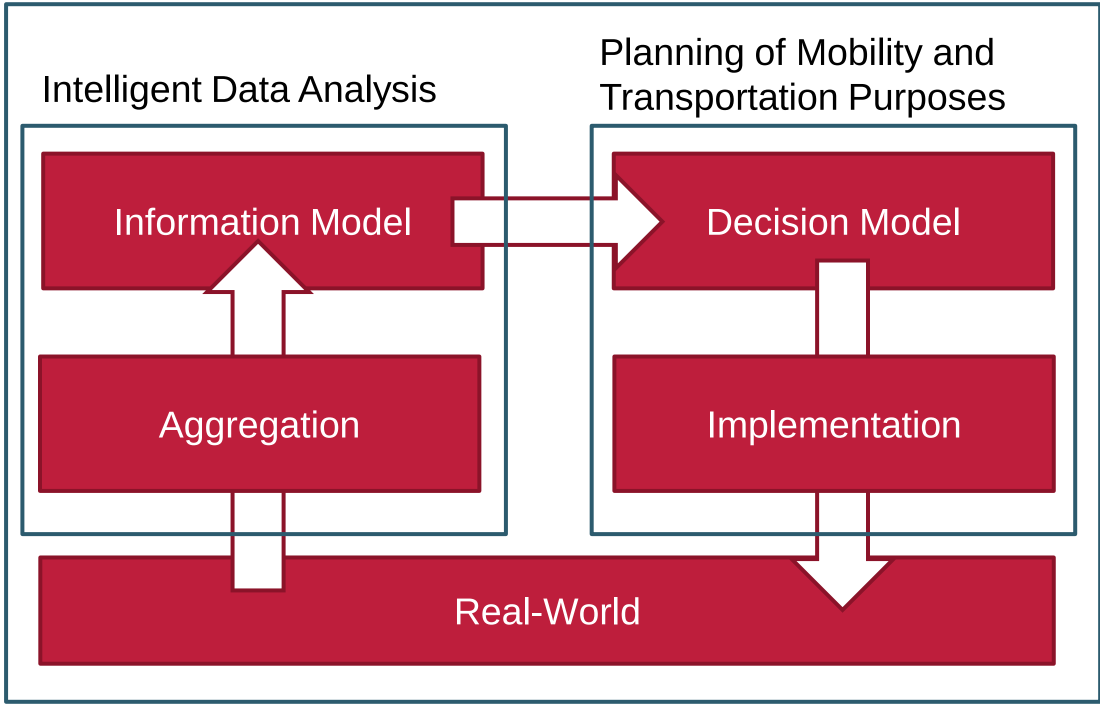
Decision making

- Data Driven Decision Making:
  - No longer elaborates on information models
  - Does not focus on optimization models anymore
  - It takes the whole loop including the real world process into account
  - This requires the modeling of the system over time → dynamic
  - The real-world may change due to external effects → stochastic
  - Implemented decisions may collide with an already changed real-world

Data Driven Decision Making



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Agenda

- Motivation
- **Operations Research & Business Analytics (Recap)**
- Modeling and Outline of the Lectures

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Example: Parcel Delivery in Braunschweig

- Truck at depot
- Customers with addresses
- Street network
- We are looking for a navigation of the driver through the city visiting every customer.
- Goal is fast delivery

- How can we achieve our goal?

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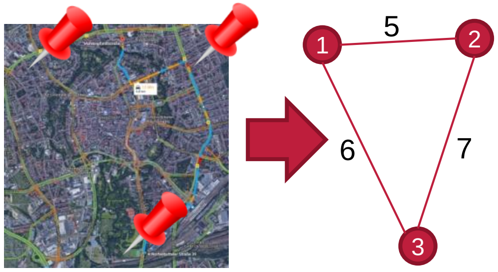
Modeling: Traveling Salesperson Problem

Part 1:

- Transfer street network to arcs between customers
- Transfer travel speeds to travel time per arc
  - → Weighted graph / Travel time matrix



Part 2:

- Transfer navigation into decision variables about used arcs
- Add constraints to ensure feasible navigation
- Define objective (with respect to Part 1)

$$\min \sum d_{ij} x_{ij}$$

subject to :

(1) $\sum_i x_{ij} = 1, \quad j = 1, \ldots, n$

(2) $\sum_j x_{ij} = 1, \quad i = 1, \ldots, n$

(3) The variables x_{ij} cannot define subtours.

(4) $x_{ij} \in \{0,1\}$

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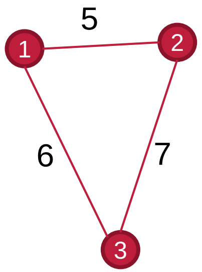
Finding a Solution

- Model spans solution space
  and provides evaluation of solutions

- Potential solutions via
  - Graph-based method (e.g., insertion procedures)
  - Mixed-integer programming (e.g., branch and bound)

- Final step: Implementation in practice (e.g., 1-3-2)
  - Translate numbers to customers
  - Map selected arcs to sets of street segments
  - Provide navigation to driver



$$\min \sum d_{ij} x_{ij}$$

subject to :

$$(1) \quad \sum_i x_{ij} = 1, \quad j = 1,\ldots,n$$

$$(2) \quad \sum_j x_{ij} = 1, \quad i = 1,\ldots,n$$

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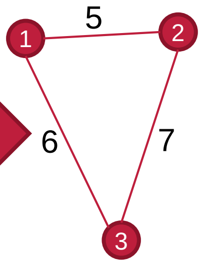
Three Steps of Operations Research

Problem definition

Modeling
(Information and decisions)

Solution
(plus Implementation)



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$(1,3,2)$

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Agenda

- Motivation
- Operations Research & Business Analytics (Recap)
- **Modeling and Outline of the Lectures**

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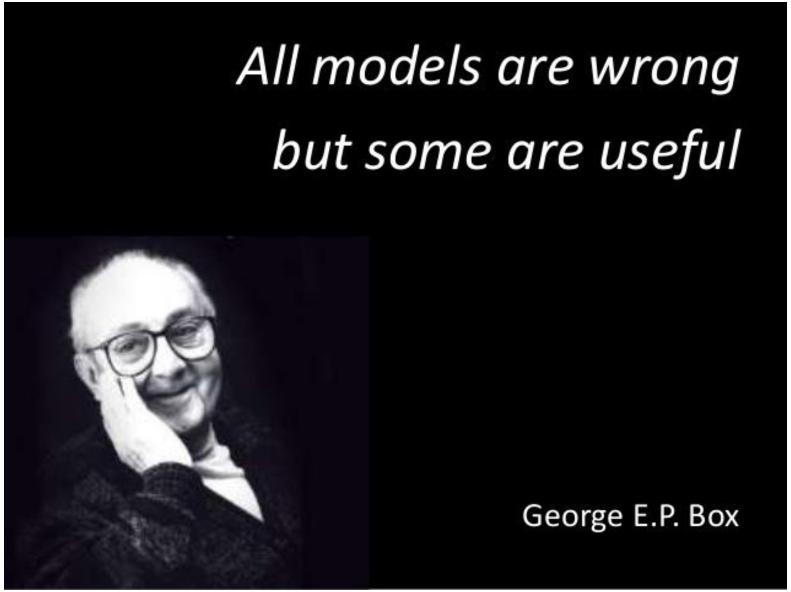
Modeling

- Different problems may share the same model (e.g., parcel delivery and technician services)

- The same problem can be modeled differently

- There is a connection between information and decision model



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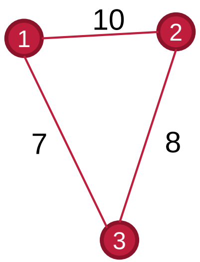
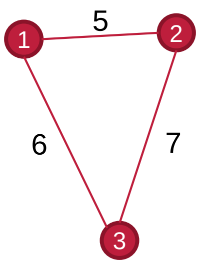
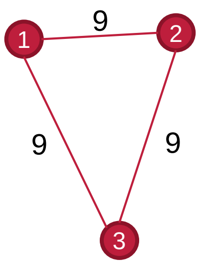
Example I: Alternative information model

- We observe daily variation in travel times, i.e., time-dependent travel times
- 7am-9am 9am-3pm 3pm-6pm







- Information model contains a set of travel time matrices
- Information generation more complex
- An alternative decision model is required (see next lecture)

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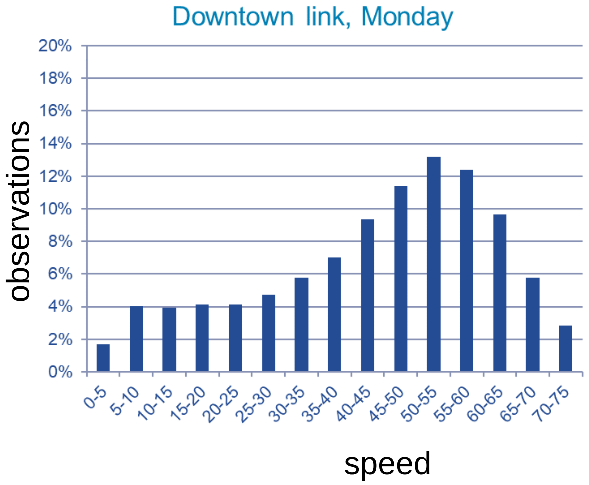
Example II: Alternative decision model

- We regularly observe overtime even though the solution "looks" feasible
- Uncertainty in travel times
- Change of decision model (stochasticity)
- Alternative and more detailed information model required (see lecture 3)



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Example III: Dynamism

- We observe changes in the information model over time
- We can react to changes (lecture 4)

- Stochastic dynamic model required (lecture 5)
- **Anticipation** of future becomes important

→ Approximate Dynamic Programming

- (lecture 6 to 10 following)

- Data Driven Decision Making in industry (lecture 11)

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subject to:

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Outlook

- Business Analytics
- Information and Decision Modeling
- Case-Study: Delivery Routing with time-dependent travel times

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